Riley Beenders

Arcane Engineer of Motion and Mayhem
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Professional summary

An innovative, hands-on, Engineer with 10 years of experience turning complex challenges into practical, scalable solutions. From designing high-precision mechanical systems to network infrastructures. I bring deep expertise in product development, CAD/CAM design, additive manufacturing, and rapid prototyping. I’ve led diverse teams, streamlined internal operations, and driven next-gen innovation in industries where precision, performance, and reliability are critical. I turn complex engineering ideas into reality by building advanced motion systems and internal frameworks that deliver precise, reliable results.

Certifications

Lean Six Sigma Certified | Green Belt

Certification completed through SSO, January 2025.

Working towards my Black Belt certification.

Education

Bachelor of Science in Electro-Mechanical Engineering | The Pennsylvania State University | State College, PA

2019 - 2024

Major GPA: *3.2*

Certificate in Mechanical Engineering | Lafayette College | Easton, PA

2018 - 2019

Relevant Experience

PROTEOR | R&D Engineer | Coplay, PA

Jan 2025 - Present

As the sole engineer in PROTEOR’s additive manufacturing department—I am responsible for the design, development, and integration of all hardware, electronics, motion control systems, and software used in our product line. Following the acquisition of Filament Innovations, my role has expanded to a global scale, focusing on innovation, automation, and product standardization across the company’s international presence.

Hobbies

- Autonomous Drones
- Homelab & Self hosting
- Hiking
- Inventing
- Machining / CNC
- Metal & Wood Artistry
- Music & Movies
- Photography
- Props and Replicas
- Robotics
- Rotary Engines / Cars
- Sound Engineering
- Take-It-Apart-ology
- Traveling

Links

LinkedIn /riley-beenders/

Languages

English *(Expert)*

German *(Beginner)*

French *(Beginner)*

Skills

Additive Manufacturing Engineer

Adobe Suite

AutoCAD

AutoCAD - Electrical

AutoCAD - Inventor

C/C++

- Linear Motor Integration and Standardization
  - Converted our flagship industrial machines—POSEIDON and ICARUS—from servo-driven ballscrew to linear motor assemblies.
  - Achieved 4 micron precision across large-format platforms with a mass of 30kg moving at 4 m/s.
  - Unified architecture between products for easier support, reduced complexity, and scalable performance across the product line.
  - Easily swappable mounts between all of our systems.
- Automation
  - Integrated a strain-gauge-based Z probing system, enabling highly accurate first-layer calibration using the machine's print head. Allowing for no user intervention—sensitive enough to detect an egg without cracking it.
  - LabVIEW - Fully automated program that gathers data from different laser sensors over a ModBus communication. Gathering nine points of data every 100ms and creating a chart that visualizes the tolerance of the data for end-customer use.
- Product Development: ICARUS-Lite
  - Designed and released the ICARUS-Lite, our most compact and purpose-built printer to date. Focused to automate the prosthetic and orthotic market.
  - Features a bolt-less shell with a seamless chamfered geometry, matching the industrial design of the POSEIDON while reducing visual clutter.
  - Pioneered a plug-and-play user experience for new customers, making professional-grade additive manufacturing accessible with minimal setup.
  - Simple electrical diagram that follows IEEE standards and is adaptable for 120/240 VAC supplies.
  - CE certification: Extensive product changes to conform to the 2006/42/EC machinery directive.
- Software Development: PROTEOR Print
  - Contributed to the development of PROTEOR Print, a next-generation remote printing interface in collaboration with an external software partner.
  - Accessible from any networked device, the interface allows full remote machine operation, diagnostics, and live video monitoring.
  - Integrated a machine vision system with AI-powered detection, alerting users to attend print failures in real time.
  - Comfortable UI tracks material usage, waste metrics, and print success/failure rates—enabling analytics-driven decision making and improved operational efficiency.
- Designed and deployed the technical infrastructure for PROTEOR's podcast and live-stream demo system:
  - Built a dual-camera + eight-microphone recording studio.
  - Developed automation software to support single-button live streaming, allowing real-time machine demonstrations for global audiences.
  - System supports simultaneous recording and live streaming for product showcases, webinars, or marketing events.

Complex Assembly Management

Computer Aided Design

CNC/CAM Programming

Customer Service

Customer/Employee Training

Data Analysis

Design for Manufacturing

Eagle

Excel

Fusion360

Fusion360 - CAM

Fusion360 - Generative Design

G-Code

LabVIEW

Lean Six Sigma

Linear Motors

Microsoft Office Suite

Network Infrastructure Design

PLC Systems

Portfolio Management

Product Rendering

Project Management

Quality Control

Rapid Prototyping/Fabrication

Self-Learning Code

Solidworks

Technical Writing

Trade Show Representation & Networking

Visio Suite

Filament Innovations | Chief Technology Officer | Whitehall, PA

Jan 2023 - Jan 2025

As CTO, I oversaw the product development pipeline, internal IT infrastructure, and strategic direction of our technology. I remain the only engineer on the team—responsible for all mechanical, electrical, firmware, and software of the systems.

- Spearheaded the transition to closed-loop servo motors, eliminating stepper resonance and significantly reducing operational noise, creating a more production-ready machine environment.
- Applied additive manufacturing to production itself—designing sleek, functional parts like custom ABS door handles that improve usability while showcasing our machines' capabilities.
- Led the full design and launch of the Gen3 Poseidon:
  - $1m^3$ build volume o Dual extrusion (filament + pellet)
  - Z-axis powered by NEMA34 motors with mechanical brakes
  - XY driven by industrial servo motors and custom gantry
- Rebuilt and expanded our IT and software infrastructure:
  - Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM system to track orders, inventory, tasks, projects, and customer interactions.
  - Automated dynamic Bill of Materials generation, linking designs directly to purchasing workflows.
- Researched and tested high-speed linear motors, achieving 3 m/s movement with a 30kg payload and 5μm accuracy—demonstrating future possibilities for precision high-speed 3D printing.
- Conducted early research into 6-axis robotic non-planar printing, achieving promising motion results and toolpath generation, though paused due to high implementation costs and recurring software fees.

Filament Innovations | Lead Electro-Mechanical Engineer | Whitehall, PA

Jun 2022 - Jan 2023

In this role, I began refining our machines beyond just functionality—focusing on manufacturability, aesthetics, internal efficiency, and advanced electrical systems.

- Enhanced thermal and acoustic management through enclosure redesign and strategic component placement.
  - Partnered with local metal fabricators to create custom tooling for large-radius bends in aluminum panels, enabling smoothed curved enclosures for improved safety and appearance.
- Integrated a new multi-board control systems connected via CAN bus, enabling simpler wiring, easier debugging, and larger-modular machine architecture.
- Developed modular electronic cases that could be assembled and tested independently, then installed into the machine in minutes, drastically improving assembly throughput.
- Collaborated closely with vendors on custom wiring harnesses, panel-cut designs, and hardware improvements to reduce manufacturing bottlenecks.

Filament Innovations | Industrial Engineer | Whitehall, PA

Apr 2021 - Jun 2022

This was a turning point in my career where I transitioned from redesigning existing systems to inventing entirely new machines and pushing innovation in large-format 3D printing.

- Designed and released Kratos, a machine specifically engineered for printing high-strength AK/BK prosthetic sockets.
- Developed next-generation models of ICARUS and POSEIDON, introducing higher-performance components.
- Designed the world's first hybrid pellet/filament gantry printer (THE KRAKEN) with a 1x2x1.5 meter build volume for a client building the world's largest 3D-printed statue.
    - Managed the full project lifecycle:
        - CAD design, scalable BOM, hardware sourcing, fabrication, assembly, and firmware tuning. In addition, I was also key in customer support and training.
- Expanded motion system safety by implementing NEMA34 motors, planetary gearboxes, auto-locking brakes during power failures.

Filament Innovations | Engineering Technician | Whitehall, PA

May 2020 - Apr 2021

With my growing knowledge and CAD skills, I took initiative to modernize and formalize the company's product designs.

- Diagnosed mechanical and electrical issues during assembly and operation, contributing to early reliability improvements.
- Gained experience tuning, calibrating, and optimizing print parameters through manual adjustments.
- Identified major design limitations in the existing machines and began documenting potential improvements.

Blue Mountain Ski Resort | Ski Lift Operator | Palmerton, PA

Nov 2019 - Jan 2021

- Operated lifts and assisted thousands of guests daily with weather info and ticketing issues.
- Quickly promoted to perform motor and control safety checks after demonstrating a strong ability to quickly understand complex electrical systems.
- Regularly called to support the maintenance team during lift breakdowns.

Northampton Area School District | Computer/Network Technician | Northampton, PA

Jun 2016 - Aug 2019

- Built network infrastructure and deployed 750+ access points across 6 buildings.
- Streamlined setup of 2,500+ Chromebooks by developing an automation script Arduinos. Significantly accelerating deployment.
- Coordinated final distribution with administration to get the devices to all the students.

RJB.Engineering | Owner & Engineering Consultant

2015 - Present

My personal company where I do a mix of everything over the years.

- Ran a small engineering consultancy delivering project-based support for automotive clients
- Launched an online store selling 3D printing heat inserts and other unique 3D printer accessories.
- 3D Printed plastic COVID face shields during the pandemic.